

Physical Action Effects in Dexterous Interaction Models: Measurement, Randomized Identification, and Task-Utility Prerequisites

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Abstract

Physical action prediction can support dexterous control only when executable interventions are informative and their information improves task decisions. We investigate these prerequisites on an existing Inspire-hand simulation substrate, independently reconstructing actuator decoding, object geometry, current-state inputs and sustained-holding labels. An initial replay procedure fails strict pre-intervention matching; subsequent prospective randomization avoids claims of cloned solver states. Several fresh action-effect and task screens fail their fixed gates. We preserve a mismatch between original 25–36-frame lift references and longer holding criteria, and separately create an explicitly synthetic 90-frame plateau task. A scratch reference-relative absolute-target actor succeeds in 56/64 trials on one preselected motion but only 56/192 overall. A 6144-trajectory randomized support forecast improves strongly against state-only prediction yet gains only 6.68% over a strong motion/action mean, failing its 10% gate. The first actual matched feature-policy experiment uses 9216 training and 1536 evaluation trajectories: all three learned variants choose identical deterministic actions on every evaluated state, and the Cm utility gate fails. Two further 1536-trajectory tests separately examine natural slip feedback and selective independent-finger corrections; both fail their prospective control gates. These results distinguish physical observability, intervention feasibility and learned policy utility. They neither establish a new general modeling method nor a universal impossibility result. A complete continuous-action auxiliary-critic comparison uses 15360 training and 1536 final-only evaluation trajectories. Its pooled physical105 successes are unchanged 137/384, cm 129/384, state only 129/384, none 147/384; the prospective Cm utility gate fails. Revision 12 adds subsequent trained-policy comparisons and a native collision- ownership correction: five shape filters were wrong under body-by-shape indexing. Earlier physics results are retained as legacy observations, not corrected-system Validation. Fixed data-scale, hand-transfer and 64/256-context studies fail their gates. Corrected-data rigid transport has positive outcome-fitted capacity, but causal learned heads miss matched gates. A full-mesh rotation screen and a fixed-label-budget disagreement acquisition study also fail their decision gates. Cm policy-training benefit remains unproved. All original failed gates, privileged-state limitations and synthetic task boundaries are retained. Validation across training seeds, generalization, hardware evidence and journal readiness remain unestablished.

1 Problem and relation to prior work

An interaction model can fit common motion while failing to compare alternative actions. Useful comparisons additionally require experiments capable of measuring physical differences at the states in which actions are selected. We examine this prerequisite alongside bounded policy-initializer checks. The repository provides self-trained actors, geometric models and a DExplore simulation substrate [2]. Historical policy audits are motivation rather than fresh evidence for this manuscript.

Shared particle actions and cross-embodiment world models already support planning [1]. Value-aware objectives [3], predictor-level action recovery [4] and multi-step controllability representations [5] are established. Contact-consistent replay and actuator residuals are also developed by Contact-UAN [6]. Our small physical MLP studies do not reproduce these systems or establish priority for their general principles. The present contribution is an exploratory evidence record with frozen hypotheses and retained failures. Recent tactile world models already forecast contact evolution: DexTacWAM models joint visual and tactile futures [8], while DexTouch-WM aligns human and robot touch/action spaces and studies synthetic policy trajectories [10].

WHIRL predicts future human-intervention probability and shapes actor risk [9]. Generic contact forecasting or predictive risk shaping therefore cannot provide a new claim for this project. These primary sources are literature context, not reproduced baselines. WorldSimProbe already formalizes action realization and grounded environment response requirements through controlled simulator evaluation [11]. Facet-0 combines action-wrench prediction, a contact-aware critic and reinforcement learning adaptation [12]. Generic physical grounding diagnostics or valuing contact consequences therefore cannot constitute our novelty. Our witness and baseline-learning tests identify experimental prerequisites, not a competing foundation model. Full methodological comparisons remain necessary.

2 Initial physical and contrast-learning screens

Four panels combine actor seeds 286/287 with environment seeds 288/289, each with 32 environments for each of three airplane motions. A retrospective trigger schedules one step after the first reference event where independent hand and object force proxies are active. A one-step wrist-z action change requests a plus or minus 10-mm PD target offset. This is a desired offset, not measured wrist travel. All subsequent normalized actions and RNG schedules are replayed. Two zero arms measure repeated process variation. Each object response subtracts its own actual pre-intervention position. Pulse and repeat contrasts are summarized by vector RMS.

Steps	Pulse RMS (mm)	Repeat RMS (mm)
1	1.914	0.000
2	2.572	0.084
5	7.150	0.101
10	18.827	0.444
20	75.463	6.081
30	105.792	9.638

Table 1: Initial fixed panels: pulse versus observed zero-repeat position contrasts.

At five and ten steps, contrast ratios are 70.44 and 42.36. At one step, the observed zero repeat is exactly zero; its ratio is undefined rather than made finite by an arbitrary denominator. Mixed vertical signs fail the joint physical gate. No beneficial lifting or population confidence bound follows from a large observed contrast.

The subsequent vector-learning decision retains this failure. It fits 192 actor-286 windows and tests 192 actor-287 windows using four past/current physical states and ten preset replay actions. Future observed physical states are absent from inputs. Matched two-layer 128-unit SiLU networks predict five/ten-step position responses; three seeds each receive 1000 Adam updates. Differential training adds measured plus-minus contrast loss. All normalization uses fit rows. Fixed test panels were already inspected by the physical experiment, making this reused-data exploration.

Input/objective	Five steps (mm)	Ten steps (mm)
Factual	10.121	26.546
Differential	10.114	26.990
No pulse	9.922	26.456
Global fit mean	10.061	26.259
Motion fit mean (privileged)	10.137	26.340

Table 2: Actor-transfer contrast RMSE, millimeters; learned rows average three optimization seeds.

Differential training does not improve over factual or global fit-mean controls. Vertical sign agreement is 32.6%/31.1%, below its fixed 75% gate. Both initial screens are UNPROMISING. Increasing steps or choosing another seed is not used to salvage these exact designs. Physical acquisition took 481.9 seconds; fitting took 17.2 seconds on one RTX 3090.

3 Geometry and larger archived predictive screen

Full sampled current geometry places 318 of 384 old triggers within 20 mm. However, actor 287’s motion 1 is entirely outside that range in both old test panels, with gaps 55–105 mm. Independent force proxies can both activate due to table contacts. The geometric shift is documented, without identifying it as the sole cause of learning failure or redefining the old gate.

We next use two immutable, episode-disjoint historical collections, 960 complete episodes each. Nonterminal frames at step modulo 16 equal to zero yield 32902 fit and 32896 test rows. These collections have been used by previous research and are not pristine Validation data. The current-near test stratum contains 1756 frames from 639 episodes; proximity uses a frozen stride-8 surface convention.

The target is one-step object-position residual over constant velocity, expressed in the current object frame. State input includes joint angles and velocities, object velocities, gravity direction, object height, palm/tip relative positions and force proxies. Matched effect networks receive either zero action slots, native commands, exact PD-target/FK palm-tip motion, predicted realized hand motion, or measured future hand motion. The last is a privileged diagnostic. An independently fit actuator model predicts joint residuals before the learned-motion variant’s FK. Effect networks have two 128-unit SiLU layers, fixed 1000 updates and three seeds. Sampling is half-near and half-all fit rows. No test-selected checkpoint is used.

Effect input	Near RMSE (mm)	All RMSE (mm)
State only	17.180	7.328
Native command	16.436	7.330
Nominal PD motion	14.599	7.093
Predicted hand motion	15.066	7.206
Measured future hand (privileged)	13.994	7.001
Passive	23.400	7.587

Table 3: Archived one-step vector prediction error, seed-mean millimeters. Near and all-state strata differ.

The predicted-motion primary method gains 8.33% over command input, failing the 10% requirement despite beating state-only, passive and its geometric transport control. Its gate remains UNPROMISING. Actuator wrist-position errors are 11.75–12.07 mm near the object. Measured future hand motion includes possible contact feedback and cannot establish a deployable causal mediator.

The separately reported nominal-motion variant gains 11.18% and is better in each seed. This motivates a separately frozen fresh causal test; it does not retroactively replace the primary method. The label-free nominal no-slip transport control is inaccurate, with near RMSE 133.64 mm. The full screen took 53.64 seconds. All 30 saved held RMSE metrics were independently recomputed.

4 Fresh proximity acquisition and failed matching requirement

New environment seeds 490/491 are combined with the same two actors. A compact reference records up to 360 steps, stopping at the first native termination. For each environment, acquisition selects the first pre-action step at or after step 10 with five remaining nonterminal transitions, both current force proxies, and full sampled surface distance at most 20 mm. Future object outcomes never choose a trigger. Frozen assets provide 1538 hand and 1024 object points.

All 384 environments acquire a qualifying state, including all three motions in every panel. The maximum sampled gap per panel is 16.60, 15.57, 18.83, 18.12 mm. Eight cold processes per panel implement two zeros and one-step plus/minus pulses in world x, y, z. Initial tensors, physical properties, actor/RMS states, actions and RNG schedules are checked. Warm solver state is not asserted to be cloneable. The existing effect models are frozen before acquisition; no refit is allowed.

The intended comparison requires pre-intervention position RMS at most 0.05 mm, maximum joint difference at most 0.0001, and maximum quaternion-component difference at most 0.0001. Joint coordinates combine translations in meters and angles in radians; quaternion differences are dimensionless component differences.

All 32 physical arms complete, but seven arms of panel actor 286/seed 491 fail the joint/orientation pre-condition. The first detected failure is zero-b, before any action intervention. Execution stops with FAILED;

Panel	Position RMS (mm)	Joint max	Quaternion max
t286_s490	0.003554	0.0000732	0.0000670
t286_s491	0.010859	0.0002204	0.0006188
t287_s490	0.007110	0.0000000	0.0000416
t287_s491	0.013398	0.0000982	0.0000881

Table 4: Maximum pre-intervention mismatch over arms relative to zero-a. Limits: position 0.05 mm; joint and quaternion components 0.0001.

the scientific transfer screen is UNCLEAR due to invalid matching. No favorable arm or window is retained, no threshold is widened, and no pooled causal score or model winner is reported. Acquisition and the failed analysis took 1152.21 seconds. This is a measurement blocker, not a refutation of the nominal predictor.

5 Exact-null recovery mechanism screen

Six native commands are overwritten by dependent finger targets. Randomizing them leaves every PD target exactly unchanged; all 32896 held rows verify this bitwise. We test whether recovering such independent null commands from model-generated object predictions induces false action responses. A truthful physical transition cannot recover an independent normalized null coordinate with population MSE below one. This is an elementary conditional-variance bound, not a novel theorem. Mutual-information sufficiency limits are already studied [7].

Every minibatch independently randomizes null commands while keeping physical labels fixed. Four matched physical predictors receive factual loss only, full inverse loss, effective-command inverse loss, or hard-canonicalized forward input with effective inverse loss. Inverse heads take current state and generated object residuals, never actual future state. Seeds 511–513 each train for 1000 updates; the auxiliary coefficient is fixed at one. Hard canonicalization guarantees null invariance by construction, without establishing learned causal correctness.

Variant	Near RMSE (mm)	Null response (mm)	Null inverse MSE
Factual only	16.557	1.118	not trained
Full inverse	16.534	1.079	1.002
Effective inverse	16.531	1.069	1.023
Hard quotient + inverse	16.564	0.000	1.023

Table 5: Exact-null intervention diagnostics, near stratum and seed means. Untrained null-head coordinates in effective/quotient variants are diagnostic only.

Full inverse null error remains near the population floor and its null response is below factual-only in every seed. Three of five interference gates fail: UNPROMISING. The proposed inverse-induced null-encoding mechanism is not supported here. Hard canonicalization removes null response, with little factual-error change. This screen took 33.06 seconds; 66 saved prediction/recovery metrics were independently audited. It is neither an AD-WM reproduction nor a general rejection of inverse learning.

6 Prospective randomized identification

To avoid invalid cold-replay matching, we acquire a single factual transition per independently assigned intervention. Four 768-environment panels combine actors 286/287 with initial seeds 492/493, 256 environments per motion. Assignment is drawn with a private CPU generator before physics. The trigger is the first even current step at or after 10 with both force proxies and full sampled hand-object gap at most 20 mm. A zero arm or one-step plus/minus wrist pulse in world x, y, z is executed once; the actor remains closed loop. All 3072 windows complete. Realized counts per motion are 37/37/37/37/36/36/36; recorded known propensities account for the unequal count convention.

Let Y be observed one-step world position residual over current object velocity. For each wrist axis j , define Z_j as the observed residual times the difference between the inverse-propensity indicators for its plus and minus arms. Under randomization and the independent-environment potential-outcome assumption, its conditional mean is the plus-minus physical effect $\tau_j(X)$. For predictions f, g , the row score is $\text{sum}(f^2 - g^2 - 2(f-g)Z)$, summing all three output coordinates and intervention axes. Its expectation is the squared conditional-effect risk of f minus that of g . This elementary estimator identifies a difference, not absolute effect RMSE or individually observed counterfactuals. Weak numeric interactions between batched environments remain an assumption limitation; exact solver-state cloning is not asserted.

Existing nominal-motion and command checkpoints 411–413 are frozen before the trial. Each scores all six candidate commands at the same current context. Per-model losses are averaged across seeds. Descriptive intervals use 2000 bootstrap replicates, preserving shared initial-seed/environment blocks across actors within six initial-seed/motion strata.

Comparison	Point (mm^2)	Central95%	Upper95%
nominal minus command	0.122	[-0.919,1.187]	1.013
nominal minus zero	-0.326	[-1.985,1.350]	1.124
command minus zero	-0.448	[-1.393,0.455]	0.296

Table 6: Randomized held-window conditional-effect risk differences, mm squared; negative favors the first predictor.

All three nominal-motion gates fail: UNPROMISING. Its historical 11.18 percent factual error advantage does not establish causal action-effect superiority. The trial takes 137.84 seconds after a retained 35.85-second device-alias engineering failure. All 3072 physical windows, risk statistics and 107 input hashes pass an independent numeric audit. We stop direct control use of these archived models.

7 Direct randomized learning and strong controls

The 3072 observed randomized windows now become fit data. A new test uses initial seeds 494/495 and all 3072 windows; no future test outcome chooses a model. This cohort changes the assignment protocol explicitly to independent seven-arm draws with propensity $1/7$, avoiding forced assignment balance. Counts are recorded as realized; no rebalancing or partial-window selection is performed.

Context comprises 63 current physical/geometry features and 18 reference commands. Six overwritten finger-command inputs are canonicalized to zero. Motion identity and observed future states are excluded. Two nuisance folds separate acquisition seeds 492/493, keeping shared actor/environment blocks together. Each nuisance model predicts Y from context, scales only its own fit fold, and predicts the other fold. Subtracting this out-of-fold response mean before constructing the inverse-propensity pseudo-contrast preserves its conditional mean. Crossfitting and randomization are established methods, not novelty claims.

For seeds 611–613, direct models have 81 inputs, 128/128 SiLU layers and nine outputs reshaped as a world contrast matrix. Factual controls receive the same context plus six pulse indicators and predict three world residual coordinates. All use Adam 0.001, weight decay 0.0001, uniform 256-row minibatches and fixed final checkpoints. Direct/factual/nuisance models get 1000 updates; the compute control gets 3000 factual updates, matching direct plus two nuisance update counts but not asserting identical FLOPs or output capacity. All scales use fit rows only. A per-seed global mean of residualized FIT contrasts and zero effect are additional strong controls.

Fresh test risk uses original observed residuals and known propensities, without nuisance fitting on test. The gate requires a one-sided 95 percent upper risk bound below zero against every control and favorable direct-minus-factual point in every seed. Bootstrap preserves the same six paired strata as above.

Direct-minus-factual seed points 32.35/19.35/20.23 all have the wrong direction. The direct method is UNPROMISING and ends without extra updates or label clipping. Fit/test take 30.31/134.49 seconds on separate admitted GPUs, producing 3.41/11.89 MiB. Six nuisance-fold scales, 16 fit model/data files, held predictions, all risk arrays and bootstrap intervals pass independent CPU/NumPy audits.

After retaining this primary failure, exploratory comparisons between the already frozen controls suggest factual-minus-zero/global differences of -3.436/-3.158 mm squared, with one-sided 95 percent upper bounds-

Direct minus control	Point (mm ²)	Central95%	Upper95%
factual	23.977	[2.024,48.106]	44.201
factual compute matched	24.278	[2.370,48.434]	44.586
global residualized	20.820	[-1.311,45.184]	40.882
zero	20.541	[-1.223,45.436]	40.638

Table 7: Direct randomized learner minus each frozen control, mm squared; every fixed gate fails.

1.106/-1.222. More factual updates have no robust incremental advantage over 1000 updates. These comparisons were selected after primary scoring; they are candidate-selection evidence, not a successful primary trial or independent Validation. A standard factual predictor is not a new method. Its candidate value is tested below using a fresh randomized closed-loop retained-success/drop experiment.

8 Randomized closed-loop task test and reference compatibility

The simple factual model’s exploratory risk signal motivates a new frozen task Probe, not a retuning of direct supervision. Equal-weight predictions from seeds 611–613 score seven candidate actions by one-step world vertical residual. Only a 10-mm wrist target pulse or zero is selected; invalid clipping candidates are masked. Controls are native actor, random valid candidate and global selection from seven FIT-only inverse-propensity mean vertical responses. No test outcome chooses their scores or correction dose.

Actors 286/287 with new initial seeds 496/497 provide four 768-environment panels. At reset, independently assign one of four policies with probability 1/4. Every policy gets the same current force/proximity gate, candidate inference and bounded opportunity budget: at most 10 events, at least six physical steps apart. Selected pulses last one step; the actor remains closed loop. Each environment is followed to its first native episode end. All 3072 episodes complete without selection.

Primary retained success requires 45 consecutive contact-supported steps at least 3 cm above initial object height, followed by no counted drop before native termination. A drop after primary success is height below 2 cm or lost conjunction of the two force proxies for six steps. The gate requires at least 3 percentage points retained gain and positive one-sided 95 percent lower bound against all three controls, plus upper 95 percent drop increase at most 2 points against actor. Intervals resample shared environment IDs across actors within six fixed strata.

Policy	Retained (%)	Stable (%)	Drop (%)	Max hold (s)
Actor	1.166	11.509	10.342	0.652
Random	1.332	10.588	9.256	0.675
Global	0.542	9.624	9.082	0.543
Conditional factual	0.803	14.004	13.202	0.815

Table 8: Randomized complete first episodes, six-stratum policy means. Stable success is 45 steps; retained additionally excludes later counted drops.

All four gates fail: UNPROMISING. Conditional-minus-actor retained difference is -0.364 percentage points, central 95 percent [-1.287, +0.575], and drop increase +2.859 points, upper 95 percent +5.478. Secondary stable-success difference +2.496 points and mean-lift difference +2.604 mm do not replace retained success. The trial takes 465.87 seconds and 44.13 MiB on one admitted GPU. Compact trajectories independently reconstruct all 3072 labels; 30440 policy choices and 142 protected input hashes pass closeout audit. No manipulation utility claim is upgraded.

Followup exposes a task-specification limitation that should have been audited before freezing the controller gate. Raw reference labels encode short periodic lifts and return to the table at the end of every motion. Full-episode nonreturn therefore asks the reference-tracking actor to depart from its intended behavior.

Eight historical native runs also have no retained success despite 70 stable instances, including correlated repeats; the new trial has small positive retained rates. Reference return does not prove the cause of every early drop. The old gate remains failed. The experiment cannot establish that all predictive control is useless or

Reference	Longest lift (steps)	Duration (s)	Final rise (mm)
s3_airplane_lift	36	1.200	0.495
s7_airplane_lift_Retake	28	0.933	0.139
s9_airplane_lift	25	0.833	0.160

Table 9: Immutable source reference labels. Every maximum continuous lift of at least 3 cm is shorter than the 45-step physical success standard.

that every counted return is accidental object loss. Repeated corrected states also differ from the first-contact contexts used to fit this model. Both task specification and closed-loop distribution need attention.

Next we freeze an explicitly synthetic holding task: duplicate 90 raw reference rows at the maximum height of each FIRST lifted interval, without selecting a pose using simulator outcomes. The native loader recomputes velocities. A new actor-only feasibility test asks for 75 consecutive contact-supported physical steps in that inserted phase, before deliberate release. Generation preserves initial rows and original remainder; it is not human-data or task-transfer evidence. Completed results of the new task are reported separately below.

9 Synthetic holding transfer and one frozen curriculum

The three first-lift label peaks are frames 72/50/62. Inserting 90 identical raw rows after each peak creates stationary phases 73–162, 51–140, 63–152, and new reference lengths 633/743/578. All 598 columns are copied together; original initial rows and remainder are preserved. The native loader recomputes all velocity channels, independently verified zero throughout the plateau.

A separate feasibility gate requires 75 consecutive physical steps within the 90-row phase with root height at least 3 cm above its actual initial height and both force proxies. It requires at least 10 percent pooled success and 5 percent in every motion. Actual native progress, rather than collector-loop indices, defines phase membership. Reference length and the complete first-episode endpoint are independently checked; no early native episode is selected.

Actors 286/287 with fresh seeds 498/499 provide four 192-environment panels, 64 per motion per panel. All 768 native episodes complete with actor/RMS frozen. The original parent reporting stage fails on a duplicate keyword; physical files are complete and preserved. A separate version uses the same frozen scoring functions and independently reconstructs every label, without rerunning physics. All 156 protected input hashes remain unchanged. Only 5/768 retained75 and 19/768 stable45 events occur; all four gates fail, UNPROMISING.

Screen	Motion	Episodes	Retained75	Stable45	Rise (mm)
Actor transfer	0	256	4	6	-10.025
Actor transfer	1	256	0	0	5.149
Actor transfer	2	256	1	13	14.190
Actor transfer	Pooled	768	5	19	3.105
PPO continuation	0	128	0	0	-1.703
PPO continuation	1	128	0	0	3.906
PPO continuation	2	128	0	0	-1.396
PPO continuation	Pooled	384	0	0	0.269

Table 10: Synthetic plateau episodes. Transfer and continuation use different cohorts and actor configurations; these rows are separate feasibility screens, not a matched treatment comparison.

One separately frozen baseline continuation starts from self-trained actor 286’s fixed epoch 420 network and both observation statistics, with fresh optimizer and new epoch count. Seed 721 receives exactly 300 new PPO epochs/460800 interactions: 96 environments, horizon 16, minibatch 512, four PPO passes, constant learning rate 0.00001. The final checkpoint alone is evaluated; no best/intermediate selection or additional optimization seed is used. Wrist-translation exploration standard deviation is predetermined at 5 mm; other coordinates retain their source values.

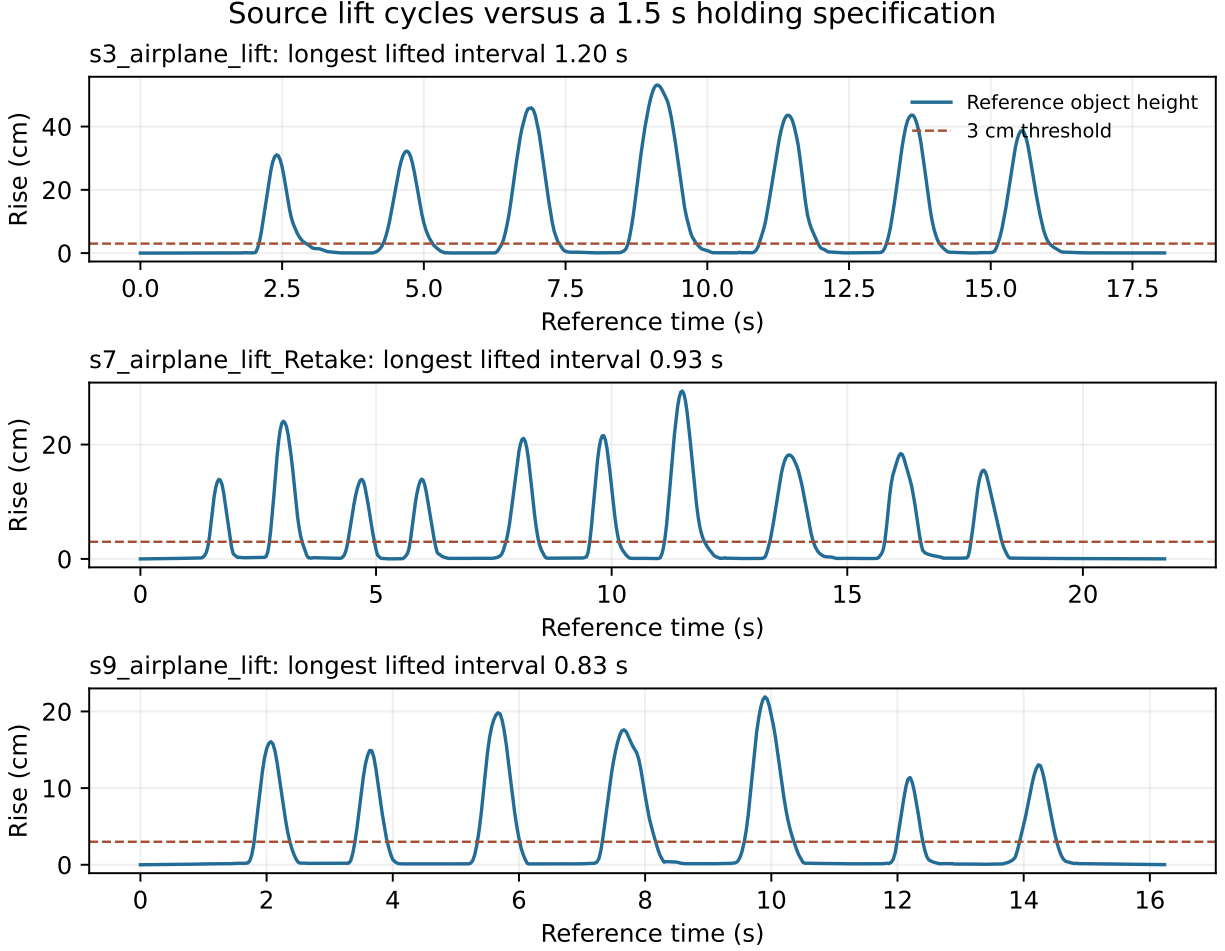


Figure 1: Source object-height targets over reference time. These are labels, not successful physical rollouts. Repeated lift/return cycles are incompatible with requiring perfect tracking and uninterrupted lift lasting 1.5 seconds simultaneously.

Half of training resets start at the first stationary row through epoch 80; this probability decreases linearly to zero at epoch 200. The last 100 epochs all start at frame 0. Native reference reward receives an additional bounded contact-supported lift bonus of coefficient 10 only during the holding phase. There is no Cm, action-selection teacher or other predictor. Training exposure is not success: 68947 plateau samples and 7933 positive bonus samples are privileged diagnostics.

The final checkpoint and evaluated model/statistics fingerprints match exactly; all weights remain finite. Fresh seeds 500/501 each complete 192 first episodes. No retained75 or stable45 instance occurs in 384 episodes; the force-proxy conjunction is absent throughout the phase. All four gates fail, UNPROMISING. All 178 protected inputs and independent trajectory labels pass closeout checks. This exact recipe ends without extra epochs, seed or learning-rate tuning. The transfer and training screens take 400.79/1120.95 seconds, respectively; one failed continuation does not establish that self-training or all curricula are impossible.

10 Native pose semantics and mechanical grip prerequisites

The actual native reset method clamps and couples finger DOFs. A three-pose label audit compares raw reference coordinates with that method applied to dummy tensors. Maximum changes are 0.29677/0.26261/0.09957 rad. Sampled surface gaps change from 0.9135/0.9456/0.4883 mm to 0.3453/1.4025/0.8457 mm; geometric finger flags remain 3/1/1. Thus reset changes do not explain failure through a demonstrated loss of these minimum-proximity or binary-flag quantities. Unsigned proximity does not establish nonpenetration or a compressive

grasp.

A mechanical diagnostic bypasses the actor: initialize at the elevated zero-velocity stationary row and hold the actual reset q as a constant absolute PD goal for 90 physical steps. Native action inversion is verified within $2.981e-8$; actor invocation raises an error and model/statistics remain unchanged. Two fresh seeds 502/503 produce 192 complete trajectories. None satisfies 75 consecutive ticks of the necessary height/proxy criterion: root at least 3 cm above original frame 0 and at most 1 cm below its elevated initial height, with both force proxies. This is neither frame 0 pickup nor proof of intrinsic grasp infeasibility.

The prospectively specified full-mesh table clearance contains a measurement error: local Z was assumed the tabletop normal. The actual 21304-vertex table is thin along local Y, and its native xyzw pose maps local -Y to world +Z. Original full-clearance metrics are therefore INVALID_TABLETOP_AXIS and cannot support a physical-distance claim. Original code, counts and failed gates remain intact. A separately recorded offline correction uses all 25002 object vertices and the actual tabletop support plane. Initial clearances are 307.03/130.93/153.56 mm. Independent NumPy world-up support agrees within 0.000000118 m; corrected retained75 remains 0/192. This is explicitly post-hoc reused-data geometry, not prospective Validation. The independently valid necessary-condition failure does not depend on the mistaken clearance axis.

A followup saved-joint audit excludes another apparent issue: wrist rotations are URDF continuous joints with no position bounds. A helper's default plus/minus pi range is not a simulator hard limit. Motion 2 joint 5 at -4.59205 rad is tracked near that value. All genuinely bounded goals satisfy their specified bounds. Nominal PD-goal agreement must also be distinguished from realized tracking: maximal wrist position deviations are 26.903/0.351/0.132 mm and finger deviations 0.0661/0.1940/0.0693 rad. These observations do not identify the cause of grip failure.

A new prospective mechanical screen asks whether additional finger closure can supply holding forces while keeping the wrist fixed. Doses 0/0.05/0.15/0.30 rad are added to independent curl joints 6/8/10/12/15; thumb yaw 14 and wrist 0-5 are unchanged. Each parent is capped by both its own and dependent native limits, then dependent coupling is reconstructed. Initial physical q is unchanged; closure is an actuator target at the first step, not a state teleport. Six overwritten command coordinates remain canonical zero. No policy is invoked and no action is clipped.

Fresh seeds 504/505 each use 96 environments. Within each motion a private generator randomly permutes eight copies of each dose before physics, giving 16 trajectories per dose/motion overall. All 192 run 90 physical steps. The full retention criterion is prospectively implemented with the correct thin tabletop geometry and independently reconstructed world-up clearance, force proxies and longest runs. Each positive dose must reach 50 percent pooled, 25 percent in every motion, and at least 25 percentage points pooled gain over zero. Candidate dose screening is not formal Validation.

Dose (rad)	Pooled /48	Motion0 /16	Motion1 /16	Motion2 /16
0.00	0	0	0	0
0.05	0	0	0	0
0.15	6	6	0	0
0.30	13	8	0	5

Table 11: Prospective bounded closure screen: retained75 counts at each dose. All dose gates fail; motion-specific counts are retained rather than replacing the pooled and every-motion criteria.

No dose meets all gates, UNPROMISING. The 0.30 rad dose meets the gain requirement, but its 27.08 percent pooled rate and zero success on motion 1 fail the other conditions. Successful instances on motions 0/2 expose a concrete mechanical candidate signal, without proving pickup, policy learning, force closure or generalization. This exact four-dose family ends without larger-dose or selected-seed expansion. The screen takes 76.99 seconds and 9.43 MiB; all 189 protected hashes remain unchanged. Independent clearance disagreement is at most 0.000000482 m. The distinct next question is grasp-pose construction and contact geometry, before another baseline learner. These statements describe the decision at that screen; the subsequent frame 0 witness and independent learners appear below.

11 Prospective support-removal witness

An original-frame 0 tracking controller follows the available reference joint plan, with a timed 0.30 rad curl increment, then holds the last plateau target. Its original strict-force gate remains UNPROMISING: unchanged/closure counts 0/96 versus 5/96 on fresh 506/507. Geometry-only motion 1 retention 32/32 is explicitly secondary. The actual object weighs 2.5936 g under native gravity; the 0.1 N net-body force threshold is 3.93 times its weight. Neither this ratio nor geometry alone establishes the hand’s contribution, motivating a prospective removal experiment.

Before new physics, motion 1 is selected as the primary stratum. In each fresh seed 508/509, 96 environments cover all three motions with 32 each. Private random assignment balances 16 continued-hold and 16 release environments per motion. All follow the same controller through the plateau stop. Release then commands open fingers and 20 cm wrist retreat; controls retain their last target. All 202 physical ticks run with gravity, unmodified object state and three native actors. The horizon is corrected from 200 to 202 before physics to cover every motion’s late window; primary motion 1 gates are unchanged. Warm solver-state clones are not asserted. Five requirements per seed are fixed: at least 80 percent prior geometry75 in each arm, 90 percent realized retreat of 15 cm, 80 percent control late retention, 80 percent release late loss, and 30 mm mean height separation. The late window is stop+30 through stop+40 inclusive, all 11 ticks required.

Seed	Prior hold	Prior release	Late hold	Late loss	Height (mm)
508	16	16	16	16	147.113
509	16	16	16	16	144.379

Table 12: Prospective primary-motion support witness. Counts retain all 16 environments per arm and seed. Prior geometry75 precedes intervention; late loss/retention require all 11 late ticks. Height is control-minus-release mean in the late window.

All ten predeclared gates pass, PROMISING as a mechanical Probe. Every release achieves measured retreat; control late retention and release late loss are 32/32. Motion 0/2 remain unsuccessful in both arms and are not excluded from the archive. Historical strictforce75 counts are 47/64 on the 64 prior geometrically held primary trajectories. This identifies disagreement for this verified setting, not a universal force-proxy error rate. The intervention provides evidence of mechanical support, not force closure or learned-policy competence. Runtime 102.35 s, 17.03 MiB, all 193 protected inputs unchanged; independent full mesh clearance error 3.80e-8 m.

12 Independent observation-policy learning

The witness motivates a new prospective holding criterion: object-root rise from actual frame 0 at least 30 mm and full mesh tabletop clearance at least 20 mm at every tick from plateau-stop-74 through stop+30 inclusive. This 105-tick window includes 75 plateau ticks followed by 30 drop-checking ticks under the holding reference. It does not relabel any historical failed gate. Strictforce 75 remains secondary. Known motion 1 must reach 50 percent pooled and 25 percent in each fresh seed; all three motions remain in training and reporting.

A random 512/256/128 ReLU/Tanh policy uses 70 features: current native joint position and velocity, current object root state, two current force proxies, 18 planned joint coordinates and planned phase fraction. Both teacher and actor have the planned reference; no observed future state, teacher action or outcome enters the actor. Seed 510 supplies all 19392 teacher rows. Fixed 2000 Adam updates, lr 0.001, batch 512, weighted action errors and final-only checkpoint use FIT normalization. The six hardware-null command coordinates are canonicalized. No source actor weights are used; it is loaded solely for native environment construction and never called. New 511/512 evaluations contain 192 complete 202-tick trajectories without fallback. Physical 105 and strictforce75 are both 0/192, UNPROMISING. Training loss is not manipulation success. Causal input and native PD reconstruction independently pass.

A separate descriptive saved-data diagnostic finds motion 1 late wrist-target RMSE 170.58/171.36 mm; input normalization does not clip any motion 1 policy context. Independent teacher-query reconstruction matches saved teacher targets. These observations motivate, without establishing a causal explanation, one fixed on-policy aggregation experiment. Fresh 513/514 collect only frozen-policy actions; teacher labels are queried offline at visited states and never executed in these cohorts. All original 19392 teacher rows and 38784 visited-state rows

are retained. No 511/512 test row is fitted. Baseline weights and normalization are preserved, optimizer reset; sampling seed 735, 2000 updates and all other hyperparameters fixed. Final-only evaluation uses new 515/516 seeds and the same physical105 gate. The aggregation also fails both gates: physical105 and strictforce75 are 0/192, including 0/64 on motion 1. The 184.70 s run preserves 217 protected inputs; this exact aggregation ends without further rounds. Smaller posthoc wrist-target errors do not replace the failed manipulation gate.

A distinct scratch actor changes the learned output to reference-relative absolute PD targets. Residual scales are 20 mm for wrist translation, 0.10 rad for wrist rotation and 0.40 rad for fingers. Targets enforce native bounds and coupling; the exact native PD inverse supplies feedback from actual current joints. Original 510 teacher rows alone train the 70-input network, seed 736, 2000 updates, unit residual MSE and final-only evaluation. No timed curl schedule executes in testing. New 517/518 produce 56/64physical105 on primary motion 1, 28/32 each seed, passing both initializer gates: PROMISING. Motions 0/2 each remain 0/64; pooled 56/192 and historical strictforce75 is 50/192. The 117.16 s run verifies all 226 protected inputs, causal context and independent residual-target/PD reconstruction. This provides an initializer, not a reference-free actor, full multi-motion competence or Cm benefit. Different losses, coordinates and physical cohorts prevent a matched superiority claim over prior learners. All earlier failures remain failed; baseline-only tuning stops here.

Policy	Motion	Trajectories	Physical105	Force75
Scratch BC	0	64	0	0
Scratch BC	1	64	0	0
Scratch BC	2	64	0	0
Scratch BC	Pooled	192	0	0
One aggregation	0	64	0	0
One aggregation	1	64	0	0
One aggregation	2	64	0	0
One aggregation	Pooled	192	0	0
Absolute target	0	64	0	0
Absolute target	1	64	56	50
Absolute target	2	64	0	0
Absolute target	Pooled	192	56	50

Table 13: Independent learned-policy evaluations: physical105 is the new prospective primary and strictforce75 is a historical secondary. All variants retain all three motions. These are initializer Probes, without Cm or policy-training utility claims.

13 Executable support prediction and strong global controls

The scratch reference-target actor is frozen. A new distribution independently perturbs initial object XY within plus/minus 10 mm before the first counted physical tick. It leaves initial height, orientation, velocity, mass and gravity unchanged. Within each motion, eight executable target changes are balanced before physics: unchanged, shared curl changes of -0.30/-0.15/+0.15 rad, or worldX/Y target changes of plus/minus 10 mm. Changes start eight ticks before reference lift and persist under the frozen learned actor. Native PD targets are recalculated each tick; commands are not numerical open-loop action replay.

An initial engineering run fails before controlled physics because a GPU state refresh immediately follows deferred reset setters. Its failed artifacts remain retained. A new frozen implementation queues native root/DOF resets and commits each combined setter once before the first simulate call. Immediate refreshes are suppressed, and every subsequent state/force write is prohibited. All 6144 new trajectories complete 202 ticks; initial packets, native targets and geometric outcomes are independently reconstructed. Full-mesh support extrema are audited using a convex hull checked against all 25002 vertices, with maximum clearance disagreement 3.20e-7 m. This corrects an implementation, not a scientific outcome.

FIT 519–524 contains 4608 rows; TEST 525/526 contains 1536. One pre-intervention row contains 69 current physical/reference features and three primitive parameters. Two matched 72-input 64/64 ReLU models share initialization 741 and 1000 Adam updates, batch 256, learning rate 0.001, final checkpoint only and FIT-only normalization. Cm receives primitive parameters; the state control receives three zero slots. A strong privileged

motion/action control estimates Beta-smoothed FIT frequencies. The primary physical30 event requires root rise at least 30 mm and entire-mesh table clearance at least 20 mm for all last 10 ticks of the next 30. It is a short physical forecast under this continuation, not a long-term action value.

Cohort	Cm log loss	State log loss	Global log loss
Motion1	0.122155	0.611316	0.130897
Motion1 seed525	0.141556	0.612498	0.142789
Motion1 seed526	0.102755	0.610134	0.119004
All motions	0.093218	0.295303	0.182906

Table 14: Frozen forecast log losses; smaller is better. Motion 1 is preselected. All-motion results are secondary and do not replace the primary gate.

Primary motion 1 has 364/512 physical30 positives. Cm improves log loss 80.02% against state-only but 6.68% against the global control, below the required 10% against BOTH. Class diversity and per-seed improvement gates pass; the primary remains UNPROMISING. All-motion physical105 is 370/1536, separately audited: motion 0/1/2 counts 0/304/66. No extra updates or subgroup replacement follow. Neither short forecast loss nor these physical outcomes demonstrate matched trained-policy utility.

14 Corrected-gradient and positional headroom screens

The next distinct cheap question uses the frozen predictors and already inspected TEST rows to study corrected gradient noise for actual physical105 reward. This is explicitly posthoc/reused-data exploration. Control variates and expected finite-action gradients are established by Q-Prop, expected policy gradients and related doubly robust policy-gradient work [13, 14, 15]. Our algebra is an application, not a new theorem. With known behavior propensity one-eighth, the sampled importance-weighted score times reward minus frozen prediction is corrected by the exact eight-action expectation of score times prediction. For any fixed prediction, the model terms cancel in expectation. Thus short-horizon predictions need not equal the longer reward to form a valid fixed control variate; that identity does not make them long-term values.

A fixed seed 751 trunk with 69 inputs and 64/64 hidden layers precedes a zero-weight 8-logit head, with bias 2 for unchanged and 0 otherwise. All 520 active head-gradient coordinates are included. At this initializer, earlier-layer gradients are zero. Coefficient 1 is fixed. Population mean gradients agree under the randomized design, so second-moment DIFFERENCES identify gradient-variance differences. Finite observed mean differences are reported without asserting population equality from the sample.

Cohort	Cm moment	State moment	Global moment
All motions	0.231475	0.284534	0.274234
Seed525	0.220441	0.259844	0.277084
Seed526	0.242508	0.309224	0.271384

Table 15: Full-gradient squared-norm means from 1536 reused rows. Absolute second moments are not themselves identified gradient variance or policy utility.

Cm lowers pooled moments 18.65% versus state-only and 15.59% versus global, failing the fixed 20% gate against BOTH. Per-seed reductions pass. The candidate is UNPROMISING and ends without coefficient or actor-seed selection. Independent NumPy reduction agrees within 2.22e-16; assigned probabilities replay exactly and finite-action cancellation tests pass. No model refitting or physical training occurs in this screen, and the preceding failed forecast gate remains failed.

A separate fixed structural benchmark asks whether current object-minus-native translation XY can change the best primitive. FIT-only medians define four bins per motion; Beta-smoothed FIT physical105 frequencies select bin-specific and global arms. There is no feature/bin scan. On the same reused TEST cohort, known randomized propensity yields Horvitz–Thompson fixed-policy estimates. These are exploratory off-policy estimates, not executed policy success rates.

Cohort	Position IPW (%)	Global IPW (%)	Unchanged IPW (%)
All motions	34.375	38.021	32.812
Seed525	33.333	40.625	32.292
Seed526	35.417	35.417	33.333

Table 16: Position-bin headroom screen. All three motions and complete test cohorts retained; reported IPW estimates do not establish executed or learned-policy gains.

The position policy is 3.65 percentage points below global and 1.56 above unchanged, failing both predeclared criteria. This exact four-bin recipe is UNPROMISING, without excluding all possible state dependence. Three failed candidate gates close the nominal eight-primitive / initial-XY family rather than motivate an unbounded local representation or hyperparameter sweep.

15 Post-lift disturbance task calibration

A new finite mechanical Probe checks whether already lifted objects have a recoverable decision under disturbance. This explicitly changes the task variant; prior physical30/105 gates stay unchanged. Initial XY is unperturbed. Seeds 527/528 retain 1536 trajectories, with 32 balanced cells per motion: four issued loads 0/0.05/0.15/0.45 N, two worldX directions and four recovery arms. Exactly one step of force is issued to the object center of mass, without torque, at plateau anchor plus 10. All other force tensor entries and ticks are zero. Object SDK body indices and tensors are independently checked. No object root or velocity reset follows initial placement. Actual contacts oppose the force; force times step divided by mass is not an observed free-flight velocity change.

One tick later, the frozen actor continues unchanged, follows the known force direction with 20 mm wrist target offset, opposes it, or applies 0.15 rad extra curl. Direction-aligned arms are privileged mechanical probes, not learned deployable policies. Prior geometry10 checks the preceding ten ticks. New recovery90 requires at least 30 mm root rise and 20 mm whole-mesh clearance on all 90 ticks from stop minus 59 through stop plus 30, allowing a bounded transient before this window. It cannot replace older failed holding criteria.

Load (N)	Prior /128	Null /32	Follow /32	Oppose /32	Curl /32
0.00	128	32	32	32	29
0.05	128	32	32	32	26
0.15	128	32	32	32	27
0.45	128	30	31	31	25

Table 17: Primary-motion mechanical calibration. Counts are actual executed trajectories; no Cm or policy training occurs. All other motions and both seeds are also reported in the retained results.

All primary trajectories have prior geometry10. Yet no nonzero load meets the required unchanged 10–80% recovery range or 15/10 percentage-point recovery contrasts. At 0.45 N, unchanged holds 30/32; following and opposing each 31/32. No load is eligible under the frozen lowest-eligible rule: UNPROMISING. The experiment completes 115.44 s, all 248 protected inputs match, and independent clearance error is at most 8.43e-7 m. End this exact force/axis/window recipe without further scans. This does not refute disturbance recovery or Cm generally. Learned dynamics planning, robust contact/motion recovery and contact-aware planning are already established [16, 17, 18]; another calibration task alone cannot supply a distinctive method.

16 Matched return-trained physical-feature policies

This is the first prospective trained-policy utility comparison on this track, rather than a forecast-loss or frozen greedy-selection screen. Code 34f78b6 freezes the scratch reference-target base actor and physical predictors. Three identical 77-input, two-hidden-layer macro policies receive current state plus eight Cm physical30 probabilities, repeated state-only probability, or privileged FIT motion/action means. All share seed752, initial

weights and probabilities, optimizer, state-only return baseline and data. Global controls execute the same neural queries before substituting FIT means.

Twelve fresh randomized panels529–540 provide9216 native trajectories. After each panel, each policy receives exactly one full-batch importance-weighted score update with actual physical105 reward, behavior probability1/8 and an ACTION-INDEPENDENT FIT motion baseline. Adam learning rate0.01, two64-unit ReLU layers, gradclip10 and12updates are fixed. There is no predicted reward, actor weight transfer from the official model, extra replay or checkpoint selection. Final-only evaluation541/542 assigns four balanced policy groups before physics; trained heads choose deterministic argmax once at the pre-lift decision.

Cohort	Initializer	Cm	State only	Global
All /384	131	151	153	140
Seed541 /192	62	79	77	70
Seed542 /192	69	72	76	70
Motion0 /128	0	0	0	0
Motion1 /128	114	113	105	111
Motion2 /128	17	38	48	29

Table 18: Fresh physical105 counts for executed policies. Each all-motion policy has384 trajectories; each seed192 and each motion128. All motions are retained. Single optimization seed: exploratory Probe only.

Cm151/384 versus state153/384 and global140/384 gives minus0.521 and plus2.865 percentage points. The frozen requirement of at least5 percentage points against BOTH controls fails, as does each-seed noninferiority: UNPROMISING. Querying all three saved head outputs on every one of1536 pre-decision states shows identical argmax actions: unchanged on motions0/1, curl+0.15rad on motion2. The tested executable decision rules coincide; unequal counts on different randomized cohorts do not demonstrate a Cm-specific advantage. This observed agreement neither proves stochastic-policy equivalence nor covers unseen states.

All14 native panel audits and12 independent NumPy score-gradient/Adam audits pass. The parent remains FAILED solely at final analysis: a25nm independently reconstructed clearance discrepancy was amplified by normalization and mistakenly checked as same-input arithmetic. Separately retained analyzerdb96433 first checks raw reconstruction, then replays actual saved native inputs with the ORIGINAL tolerances. No resimulation, retraining, outcome edits or threshold change. Corrected maximum raw/feature/logit errors are2.49e-8m,1.06e-6 and7.52e-6; actual macro argmax agrees exactly. All374 parent input hashes are reverified.

This is established frozen-feature contextual policy learning, not a distinctive algorithm. Physical predictor pretraining4608 trajectories and base teacher19392 rows are reported separately from9216 policy-training trajectories; no total sample or compute saving is claimed. Stop this exact recipe without additional updates, learning-rate/head/seed selection or altered deployment. Future work must establish physical decisions beyond a per-motion constant action before investing in another policy-training comparison.

17 Natural support loss and prospective feedback feasibility

A separately labeled training-only diagnostic retains all1152 unchanged-arm trials from twelve randomized panels529–540. Motion0never acquires five consecutive lifted ticks; motion1has56/384 and motion2has101/384 cases that acquire then fail physical105. Every such case shows object-minus-hand downward speed before first loss. Five-tick acquisition is only a diagnostic, never a replacement success criterion; no evaluation data or learned model is used.

Code7e2c741 freezes a new prospective native experiment543/544. All1536 trajectories retain original initial-XY variation, reference-conditioned scratch base,202ticks, material/mass/gravity and physical105. Four privately preassigned policies are unchanged, early curl+0.15rad at lift_start-8, event curl+0.15rad, and event wrist-XYZ arrest. Before each action, a current-state trigger latches five consecutive physically lifted ticks from lift_start, then fires once while still lifted when object-minus-hand vertical velocity is at most-20mm/s. Event curl persists; wrist arrest holds the measured translation at trigger while orientation and fingers continue the base policy. No object/hand state writes, external forces or future outcomes enter this controller.

Event curl92/384 trails unchanged133/384 by10.677 percentage points. Wrist arrest125/384 trails unchanged by2.083 points, although it exceeds harmful early curl96/384. Neither satisfies the predefined5-point gain over

Cohort	Unchanged	Early curl	Event curl	Wrist arrest
All /384	133	96	92	125
Seed543 /192	68	48	40	65
Seed544 /192	65	48	52	60
Motion0 /128	0	0	0	0
Motion1 /128	111	67	71	107
Motion2 /128	22	29	21	18

Table 19: Prospective natural-retention physical105 counts. Each pooled policy has 384 trajectories, each seed 192, and each motion 128. All trajectories and the original criterion are retained; no event-conditioned subgroup replaces the primary. These are mechanical controller results, not Cm training.

BOTH controls or each-seed noninferiority: UNPROMISING. Complete independent NumPy event timing, current state, native target/PD, private assignment and full-mesh labels pass; maximum clearance error 2.607×10^{-7} m. Run completes 114.312 s, 271742088 bytes; all 251 protected input hashes are rechecked. Stop the exact trigger/action family without threshold, dose, axis or timing scans.

Observability does not establish intervention value in this setting. Unchanged motion 1 has 82 successful and 17 failed trajectories that trigger; motion 2 has 10 successful and 39 failed triggering trajectories. Relative downward speed need not imply that more closure or wrist arrest helps. Early curl changes the pre-event physical prefix, so event-conditioned cross-policy rates are not matched causal comparisons. Root pose/velocity and joint state are privileged simulator observations; no sensing or hardware transfer is demonstrated.

Slip-direction grasp reflexes are already established[19]. Predictive and reactive tactile residuals[20], action-conditioned pre-contact failure monitoring[21], and causal temporal detectors with conformal alarm calibration[22] also precede this screen. A generic forecast-plus-feedback design is insufficient for a distinctive method. The next decision concerns independent effective finger actions: common curl can harm successful contacts, while a visible loss cue does not specify a useful correction. Selective-actuator feasibility still would not establish Cm utility; a subsequent method requires matched actual policy learning and Validation.

18 Selective independent-finger feasibility

After the failed slip-reflex screen, we test whether the common curl synergy conceals useful independent corrections. The native SDK and the read-only Inspire URDF agree on six effective coordinates: index 6, middle 8, pinky 10, ring 12, thumb yaw 14 and thumb curl 15. Children 7/9/11/13/16/17 are reconstructed from parent commands and are not independent policy dimensions. This action accounting is an implementation prerequisite, not a new modeling method.

Two fresh seeds, 545 and 546, provide 1536 native trajectories of 202 ticks each. Before physics, eight arms are independently randomized within each motion: 32 trials per arm, motion and seed. We retain the same base actor, references, private initial XY offsets and physical105 criterion. At eight ticks before lift start, we persist either no correction, shared curl ($+0.15$ rad on coordinates 6/8/10/12/15), or a single $+0.15$ rad correction on one of 6/8/10/12/15/14. Wrist targets are unchanged. Requested offsets and actual bounded target offsets are recorded separately; dependent and native limits are independently reconstructed. The preselected motion 2 gate requires at least 20 percentage-point gain over both unchanged and common curl, with no loss to either control in each seed. The six selective candidates constitute a declared mechanical screen, not a fitted policy or formal validation.

Motion 2 gives 12/64 successes for the unchanged actor and 21/64 for common curl; selective arms yield 4–10/64. All primary candidate gates fail, UNPROMISING. The run completes in 112.349 s and retains 279708481 bytes; all 255 protected inputs and full-mesh, current-state, native-PD and assignment audits pass. Independent clearance error is at most 0.000222 mm. This closes the exact positive-offset family without sign, dose or timing scans. It does not rule out continuous feedback learning, negative adjustments or other task distributions.

At the first intervention in motion 1, requested thumb yaw is target-null in 64/64 trials and thumb curl in 47/64; both encounter bounds in 64/64. These current command/limit diagnostics are not future measured hand motion, whole-trajectory counterfactuals, or evidence that unused dimensions can never matter. Independent randomized cohorts do not identify interaction effects between combined fingers.

Arm	Motion 0 /64	Motion 1 /64	Motion 2 /64	All /192
Unchanged	0	54	12	66
Shared curl	0	39	21	60
Index	0	39	10	49
Middle	0	45	6	51
Pinky	0	55	10	65
Ring	0	56	6	62
Thumb curl	0	54	8	62
Thumb yaw	0	53	4	57

Table 20: Fresh selective-finger physical105 counts from all 1536 trajectories. Each motion has 64 trials per arm; pooled counts have 192 per arm. Motion 2 is the preselected primary. Every fixed selective-correction gate fails.

19 Complete continuous policy-training comparison

We freeze a 12-coordinate continuous PPO comparison: Cm physical auxiliary critic, state-only auxiliary control and no-auxiliary control, all with identical initial actor and critic weights, 20 updates and equal native interactions. The value baseline is state-only. Cm receives current state and executed target minus current joint position, and predicts actual next object position and linear-velocity changes. Separate actor and critic encoders prevent direct auxiliary gradients to the actor; joint gradient clipping can nevertheless couple the actor update scale to critic gradients. The only reward is actual terminal physical105, never a model prediction or surrogate reward.

Gaussian request likelihood is computed before tanh and native projection; it is not executed-target density or entropy. Auxiliary actor-critic objectives are established [23], and clipped-tail score correction overlaps CAPG [24]. This recipe is not a new general method.

Training uses seeds 547–566, 768 independent randomized native environments per panel, 192 per arm across three motions. All three learned variants complete 3840 training episodes and 3040 PPO minibatch updates each, totaling 9120 optimizer steps. Evaluation uses only the final checkpoint, deterministic actions and preselected seeds 568/569. The engineering seed 567 never enters training or evaluation. The primary gate requires pooled Cm success to exceed BOTH learned controls by five percentage points, Cm to be no worse than unchanged pooled, and Cm to be no worse than both learned controls on EACH evaluation seed. All seven conditions are required; no subgroup can rescue a failed gate.

Cohort	Unchanged	Cm	State only	No auxiliary
Pooled /384	137	129	129	147
Seed 568 /192	68	59	65	71
Seed 569 /192	69	70	64	76
Motion 0 /128	0	0	0	0
Motion 1 /128	114	109	108	118
Motion 2 /128	23	20	21	29

Table 21: Final-checkpoint-only physical105 success counts. Denominators in each row apply to each arm. Motions and environment seeds share one optimization seed; these are Probe results, not independent training replications.

Pooled counts are unchanged 137/384, cm 129/384, state only 129/384, none 147/384. Cm minus state-only is 0.000 percentage points; Cm minus no auxiliary is -4.688 percentage points. The fixed full-coordinate auxiliary-critic recipe is UNPROMISING: it fails 6 of seven prospective gates. This exact recipe stops without coefficient, learning-rate, seed, checkpoint or training-duration searches. It does not refute other uses of physical interaction models.

All 22 native panels are independently reconstructed from their raw trajectories, including current-state inputs, request probabilities, executable targets, native PD and complete-mesh physical105 labels. All actor and critic gradients and Adam updates for 60 predetermined first minibatches are checked independently in

NumPy; the other 9060 optimizer steps are retained and counted but not independently replayed. Full coverage and all seven final gates are separately reconstructed with integer success counts.

The original update-17 NumPy audit fails its frozen gradient threshold at one first-layer ReLU: float64 and GPU float32 arithmetic give opposite signs within roundoff of zero. A saved-batch GPU replay reproduces every saved gradient, forward output and loss exactly, without optimizer or simulation updates. A separate NumPy audit uses the witnessed branch only after independently checking the FP32 dot-product rounding enclosure. All ORIGINAL scalar tolerances then pass, including maximum gradient discrepancy of $1.223\text{e-}6$. The original failed audit/parent remain unchanged, and the correction is recorded separately. Training resumes from the exact saved update-17 Adam state; no valid optimizer update is repeated. The audit is explicitly branch-aware, not a claim that float32 and float64 derivatives agree at nonsmooth boundaries.

After an execution interruption, completed panels and full Adam checkpoints are retained exactly. A subsequent native allocator failure on GPU1 is preserved, alongside one engineering-only reproduction excluded from training and testing. The remaining phases run on an admitted idle GPU4 of the same RTX3090 model; this is an explicitly recorded runtime migration, not uninterrupted execution on a single device. No valid panel or optimizer update is repeated or selected. Initial reset packets match across the failed and recovered panel, without a claim of cloned solver state. All controls share the device within each panel.

Physical105 means root rise and complete-mesh table clearance on all 105 specified ticks. It does not establish force closure or mechanically supported grasp in every successful learned-policy trajectory. An earlier randomized support-removal witness concerns a different base-policy cohort; it is not automatically transferred to the learned policy.

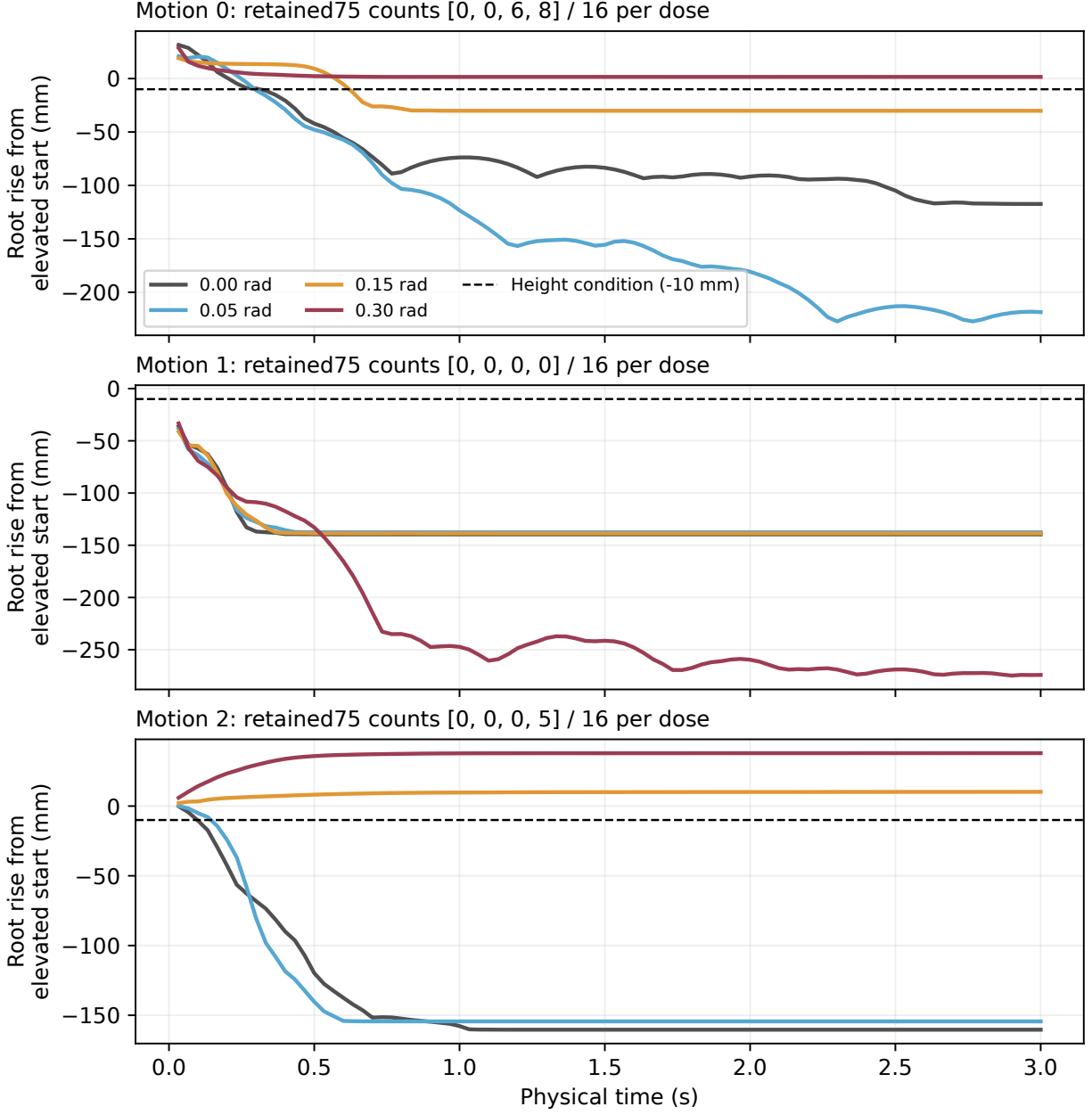


Figure 2: Physical object-root trajectories for the prospective closure screen. Each curve averages 16 trajectories per motion/dose. The dashed line is only the initial-height condition; forces, table clearance and 75 consecutive steps are additionally required. Mean height is not retained success or a policy benefit.

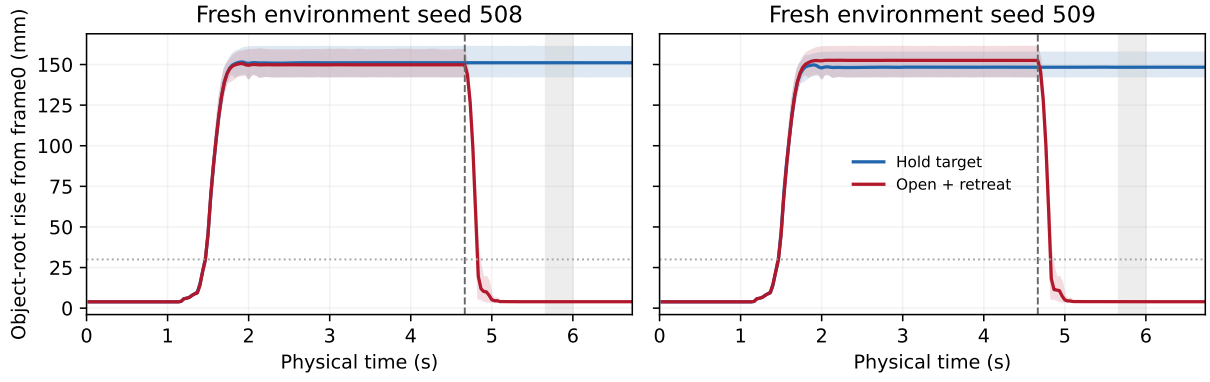


Figure 3: Actual primary-motion object-root trajectories, all 16 assigned rows per curve. Bands show minimum/maximum across trajectories, not confidence intervals. Dashed vertical line marks intervention; shading marks the 11-tick late window. No learned actor controls these trajectories.

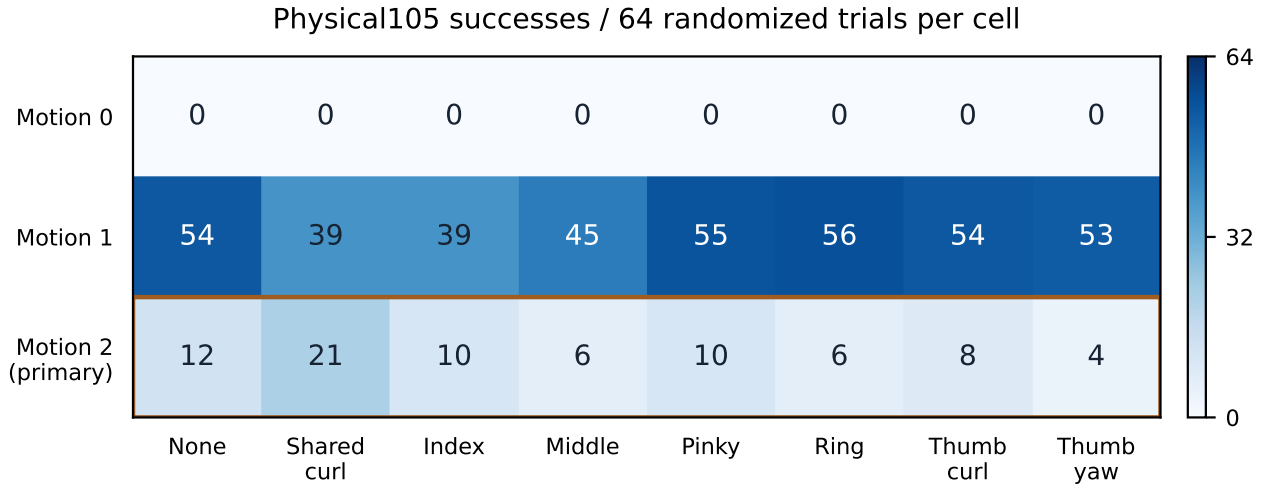


Figure 4: Recorded physical105 successes for all eight prospective arms in every motion. The outline identifies the preselected primary. This fixed mechanical screen measures no learned Cm policy performance, and no subgroup replaces the failed primary gate.

20 Subsequent evidence and corrected physics

This revision integrates later completed Probes while retaining every original failed criterion. Unless explicitly marked corrected below, the historical policy and randomized-effect results use legacy simulation filters. They cannot be treated as corrected-physics Validation. The task is one airplane with three synthetic references, not the user workspace’s separate expert/motion results. All listed policy experiments use self-trained actor parameters; pretrained behavior in early data-source diagnostics is not evidence of this objective.

20.1 Later policy-training comparisons

Deterministic successor composition, an observed atomic successor distribution, physical-feature-to-Q learning and equal-interaction-budget auxiliary learning all fail their complete utility gates. Counts below are separate fixed recipes, not a pooled effect estimate. Per-recipe environment seeds share one training initialization and do not provide independent training-seed Validation. Cm’s occasional advantage over one control cannot replace the all-control gate.

Recipe /384	P0	Cm	Off	Task Q
Deterministic successor	129	172	174	222
Observed successor distribution	126	188	188	222
Physical encoder to Q	143	219	218	225
Cold-Q budget block	129	200	198	189
Equal-interaction budget block	115	183	201	220

Table 22: Later trained-policy physical105 counts, each arm /384; legacy physics. The task-Q column in the last two rows uses different declared budgets.

20.2 Collision ownership correction

Native SDK inspection finds 25 rigid bodies but 13 collision shapes. Indexing body names by shape number misassigns five shape filters. Actual SDK ownership fixes non-thumb shapes 2/4/6/8 from 2 to 3 and thumb-base shape 9 from 3 to 2. The native check has 5 mismatches before and 0 after under the same URDF. One frozen-policy panel is independently rerun, with full geometry/command/label audits. Subsequent representation

Physics /192	P0	Cm	Off	Task Q
Before ownership fix	65	96	101	95
After ownership fix	61	81	92	91

Table 23: Fixed-policy sensitivity to corrected collision ownership. One seed, no retraining, no exact paired solver states. PROMISING refers to sensitivity, not Cm utility or performance improvement. All motion 0 counts remain zero.

studies reuse this corrected655 panel:384 training and 384 held episodes with a fixed whole-episode split. The holdout is repeatedly viewed; it supplies route-selection evidence, not fresh Validation.

20.3 Data scale, hand morphology and query granularity

The source surface caches contain 2048 MANO and 10135 Inspire hand points. The initial prior uses 64 object queries with four averaged hand neighbors per query. Its 22 features include realized future hand motion; offline accuracy is not a deployable command-conditioned forecast. Fifty training objects perhand and nested 512/2048/7168 windows provide the scale comparison. Parent-equal EPE is used, without selecting checkpoints on evaluation data. Pure hand morphology is not identified because data distributions and simulation provenance differ. The initial-to-largest own-hand gains are 8.015%MANO and 6.560%Inspire, below the 10%gate. Fixed

Training windows / adaptation	MANO / prior	Inspire / scratch
512	3.869332	4.743223
2048	3.601316	4.416893
7168	3.559204	4.432058
256-label adaptation	4.726920	4.830451

Table 24: Own-hand prediction EPE(mm), except the final row: MANO 7168 prior after fixed 256 Inspire labels versus scratch on the same labels. That row evaluates Inspire for both columns and is not an own-hand scale comparison.

256-label adaptation improves 2.143%against scratch, below 10%, and 2.330%against shuffled pretraining, below 5%. The granularity experiment fixes 2048 training windows perhand, common initialization,23107 parameters and 1500 updates perfit, changing 64/256 context and mean/detail neighbor features. All supervised targets remain 64 queries; this is not increased target resolution or a FLOP-matched comparison.

Context / neighbor aggregation	MANO EPE(mm)	Inspire EPE(mm)
64 / mean	3.456930	4.392716
256 / mean	3.457202	4.390056
64 / detail	3.465854	4.399568
256 / detail	3.464157	4.397399

Table 25: Fixed-data own-hand granularity comparison; all six 10%gates fail. These are eight fits, not independent training-seed replications.

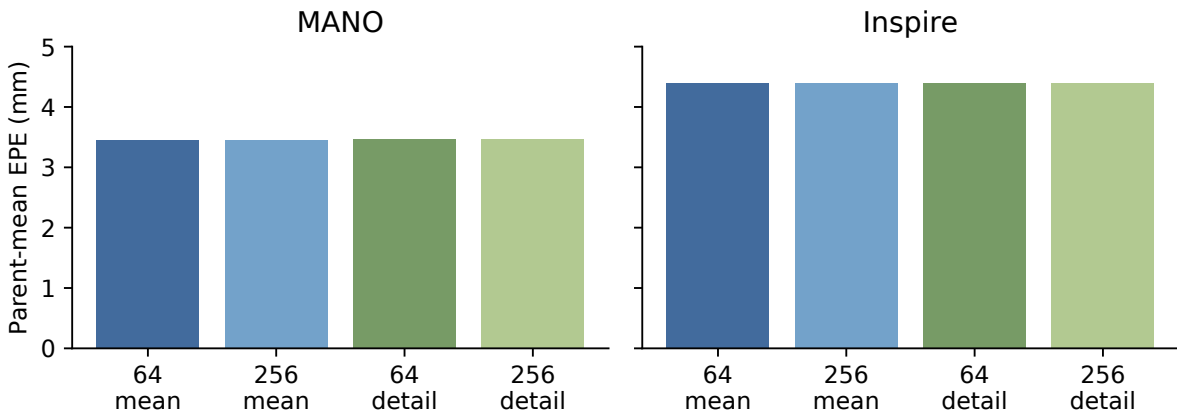


Figure 5: Recorded parent-equal errors with an untruncated zero-origin axis. No uncertainty bars are invented from a single optimization seed.

20.4 Corrected causal input and failed transfer calibration

A 54-coefficient train-only actuator fit reconstructs next hand geometry from current state and commands, yielding 0.900 mm hand EPE against 2.601 mm velocity persistence. This qualifies an execution bridge on the reused panel; it does not establish hand-contact attribution or policy utility. Frozen priors fail even with measured next hand motion. Calibration uses four matched 1200-update heads and fixed final checkpoints; only the hand-flow-removal contrast passes. An independent calibration audit initially fails from NumPy indexing

Forecast variant	EPE(mm)
mano / oracle	7.301355
mano / action velocity	7.346440
inspire / oracle	6.725405
inspire / action velocity	6.753861
Calibrated / pretrained	0.665444
Calibrated / scratch	0.633220
Calibrated / shuffled	0.618116
Object-flow persistence	0.629373

Table 26: Corrected-panel object-flow EPE(mm),6144 windows/384 held episodes. Oracle rows use realized future hand motion and are nondeployable. Calibration and frozen-prior variants are different models, not one ablation matrix.

axis order; the separate audit-only recovery retains all fitted weights and labels without repeating any optimizer update. Protected raw packets and full held forward/metric audits pass.

20.5 Rigid transport capacity versus causal learnability

The geometric capacity screen constructs 13 hand-link transport endpoints plus zero/inertia. Outcome-fitted selection and scalar weights minimize actual held errors on the union of 15 segments. It is not the full convex hull or a physically feasible contact law. Its 20.173% advantage over outcome-fitted state control is PROMISING capacity only: weights and endpoint choices use future outcomes. Causal coefficient/score fitting then fails matched learning gates. A separate state-anchored correction expands the mixture family and directly optimizes flow EPE, preserving a frozen state baseline outside a hard current 2 cm proximity gate. Its 8.955% gain misses the fixed 10% matched-state gate. Far-field preservation is a hard rule, not learned generalization. Initial state-

Different model families	Full EPE(mm)	Matched state(mm)
Outcome-fitted segment oracle	0.294084	0.368403
Learned coefficient / score	0.558129	0.448096
State-anchored convex correction	0.395165	0.434032

Table 27: Different transport families, equal-episode EPE(mm) on corrected held data. Each row has its own matched state control. The first row is an oracle; it must not be read as a deployable winner over the learned rows.

anchor fitting fails before any head update because CPU normalization changes the original GPU arithmetic; the repair restores the exact frozen baseline and retains the original tolerance. The successful run completes 4500 new head updates, separately from 1500 inherited state-model updates. All held neural outputs/metrics pass; 9 early AdamW updates are independently replayed, 4491 remaining new updates are counted but not independently replayed.

20.6 Rotational target adequacy

An older six-dimensional auxiliary target omits future rotation, whereas the 64-point flow already contains it. An observed geometry decomposition evaluates ALL 155904 poses and 155136 transitions with the 25002-vertex mesh and a fixed table upper plane. Current-only eligibility requires 30 mm root rise and 15–25 mm clearance. The true-future-translation diagnostic persists CURRENT orientation separately at EACH step. It is not a simulated no-rotation trajectory or causal intervention. All 768 full105 task labels agree between actual poses

Split	Windows / episodes	Flips(%)	Rotation95th(mm)
train	148 / 123	2.913279	8.502605
held	144 / 113	1.917404	7.039685
all	292 / 236	2.436441	7.685484

Table 28: Equal-episode decision flips and weighted 95 th absolute rotational clearance contribution. Held coverage and 2 mm magnitude gates pass;10%flip gate fails. Motion 0 has no eligible windows, so it was not assessed.

and this diagnostic, with 325 successes (154/384 held). CPU SciPy/NumPy independently rebuilds ALL mesh poses and transitions; mesh differences are 1.11×10^{-16} m versus GPU and 5.61×10^{-8} m versus retained native clearance. These observations close rotation alone as the current remedy, not longer-horizon rotation or other geometries.

20.7 Physical-response uncertainty for label acquisition

World-model-guided exploration[25], structured curiosity for object manipulation[26] and tactile/contact-prioritized replay [27] are established. We test a specific offline mechanism rather than claim generic exploration novelty: bootstrap disagreement of plus-minus physical responses, which cancels common per-model state-dependent offsets. This algebra does not prove calibrated epistemic uncertainty or useful labels.

Legacy randomized FIT 492/493 contains 3072 windows. Three 600-update bootstrap models use 512 initial labels, preserving shared actor/environment blocks. Each acquisition method selects 512 additional labels using only current contexts, hypothetical actions and frozen models. Three final 87–64–64–3 MLPs share initial weights, initial-only scales and 1000-update schedules. Total 4800 new updates; no native simulation or actor updates. The independently acquired IID TEST 494/495 windows were previously viewed; they are not new Validation.

For physical response Y , randomized assignment A with probability p , the pseudo contrast for axis j is

$$Z_j = Y \left(\frac{\mathbf{1}(A = 2j + 1)}{p_{2j+1}} - \frac{\mathbf{1}(A = 2j + 2)}{p_{2j+2}} \right), \quad j = 0, 1, 2.$$

For fixed predictors D_a, D_b , relative squared causal-effect risk is identified by the mean of $\|D_a\|^2 - \|D_b\|^2 - 2\langle D_a - D_b, Z \rangle$. Absolute effect RMSE is not identified. Two thousand descriptive bootstrap draws preserve shared actor/environment blocks within test-seed/motion strata. The uniform-comparison points are +0.069200

Contrast minus comparator	Risk(mm ²)	One-sided 95% upper
uniform	0.362847	2.202990
absolute	-1.923996	-0.126221
zero	-2.988855	-0.446212

Table 29: Action-contrast-disagreement acquisition versus uniform acquisition, absolute-response-disagreement acquisition and zero-effect prediction. Lower relative risk favors contrast acquisition. Only 3/6 gates pass: advantage over absolute disagreement cannot rescue failure against uniform acquisition.

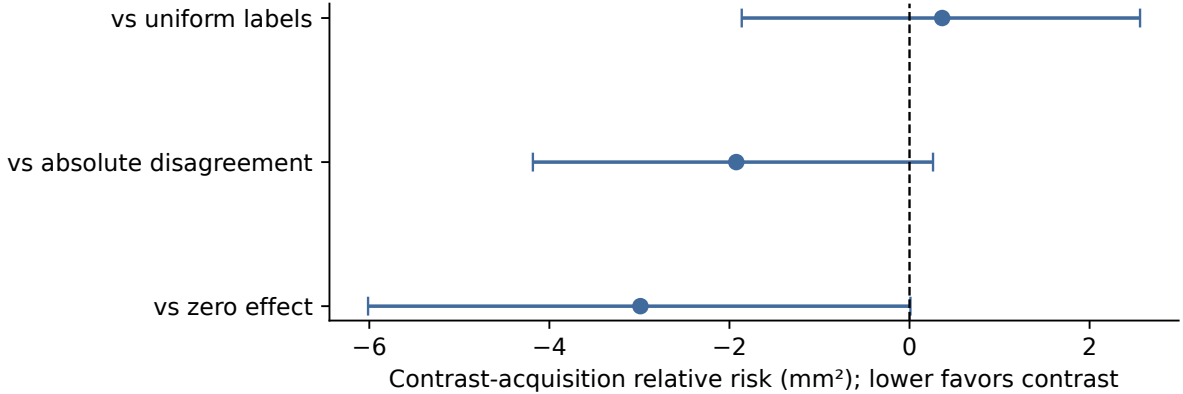


Figure 6: Recorded point estimates and central 95% descriptive intervals for fixed fitted models. One-sided 95% upper bounds in the table use a different quantile; the negative upper bound against absolute disagreement does not make its central 95% interval exclude zero.

mm² on 494 and +0.656493 on 495. The complete gate is UNPROMISING. All 6144 raw responses/assignments, every new forward/rank/selection/metric/bootstrap and 18 early AdamW updates independently pass, with 5.97×10^{-9} m forward discrepancy; 4782 updates are not independently replayed. The actual run costs 22.10 s/9.92 MB. All 3072 FIT pool and 3072 TEST windows were already paid; 1024 selected labels must not be called 1024 online interactions. The native acquisition route is stopped without further seed/budget rescans.

21 Limitations and research decision

These experiments involve one object, three shared motions, two existing actors and small fixed fitting procedures. Contact proxies plus sampled proximity are not attributed pairwise forces. Short-horizon measurement is not stable lift, retention or manipulation utility. One fixed baseline PPO continuation is presented; The matched frozen-feature comparison is negative; the continuous comparison is a completed single-seed Probe; unseen-object/hand evaluation and hardware deployment remain absent. Optimization seeds do not create independent physical datasets. Existing papers’ claims are not replaced by these diagnostic tests.

The direct learner and tested greedy controller are stopped at their fixed gates. The reference-target initializer supports bounded physical studies, but the subsequent forecast, gradient, position and finite-disturbance candidates fail. Matched Cm-on/off policy training has been executed in two distinct fixed designs; meaningful conditional decision headroom is the consequential next requirement. A compatible reference is a prerequisite, not proof of controllable contact or model benefit. Synthetic plateau transfer, one fixed continuation and mechanical feasibility screens are completed with their original failed gates retained. Corrected geometry is labeled explicitly as a separate reused-data diagnostic.

The original user objective remains a journal-level research outcome. This working paper does not meet it: a distinctive method and consequential matched task evidence remain necessary. All negative outcomes and the bounded continuous Probe are retained to guide further work rather than converted into positive conclusions.

Reproducibility

Current independent work is in `agent/cm-active-acquisition`; earlier research branches and all historical revisions are retained. Original checkpoints, motion assets and prior evidence are read-only. Physical/differential code is fixed at `466e80d/44f22ea`; factorization at `bdb5ba4`; fresh collection at `59de1cf`; null recovery at `33d1085`. Randomized collection uses `510a00e`; direct fitting `2075314`; independent auditing `f283bb9`. Closed-loop task collection uses `c0ec955`; task/reference audit and next synthetic task design use `b2b26e0`. Holding collection/corrected reporting use `40e73f2/3f4ae77`; the frozen continuation uses `e770d4b`. Native reset/static diagnostics use `da39660/1e38cf6`; the independent geometry correction uses `68bf178`; prospective finger closure uses `5d7c375`. Frame 0 tracking uses `d15ef0f`, support witness `49cf8b3`, scratch observation policy `9c0a218`, and fresh corrective-label aggregation `e2a38c2`, reference-target actor `596c5a0`. Executable support forecasting/reset correction use `5140208/3ac4f6c`; corrected-gradient screen `ef71c33`, positional screen `f66aa01`, and one-step physical disturbance calibration `721b44d`. Manifests preserve commands, input SHA, GPU UUID, budgets and failures. The first feature-policy comparison uses `code34f78b6`; its separately retained final audit correction uses `db96433`. Every table and figure comes from recorded labels or physical trajectories, clearly distinguished. Revision 12 is compiled from native LaTeX. Earlier ReportLab review copies, their exact inputs and the failed revision-10 native compilation are retained. Earlier manuscript revisions are preserved separately.

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